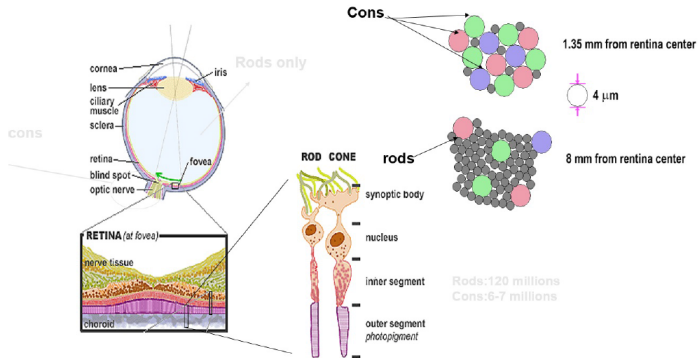


Lecture 2 Transform

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Department of Electronic and Computer Engineering

HVS

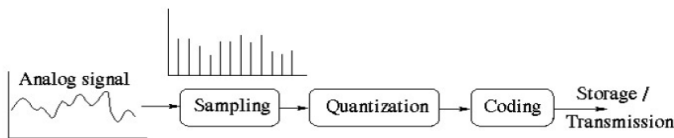


*Cones: luminance (bright) and color

*Rods: luminance only, but 10,000 times sensitive to cones

Analog to digital conversion

- ▶ Sampling: Sampling frequency should satisfy Nyquist Criterion
- ▶ Quantization



Sensor

Also quantization of amplitude

Sampling and Quantization of Digital Images Using CCD Cameras

- ▶ In CCD camera, sampling and quantization are key elements
- ▶ **Sampling:** Picture sampled by CCD chip grid
- ▶ **Quantization:** Each CCD element uses fixed number of bits to represent image values
 ⇒ We represent an image as a matrix of numbers
 ⇒ MATLAB is the ideal tool of image processing

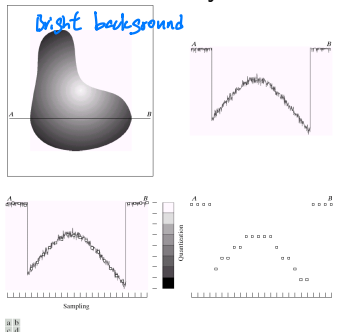
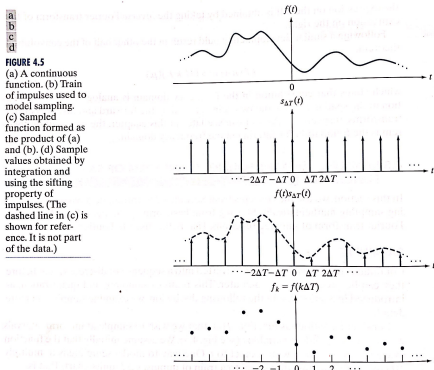


FIGURE 2.16 Generating a digital image. (a) Continuous image. (b) A scan line from A to B in the continuous image, used to illustrate the concepts of sampling and quantization. (c) Sampling and quantization. (d) Digital scan line.

Effect of Sampling in the Spatial Domain

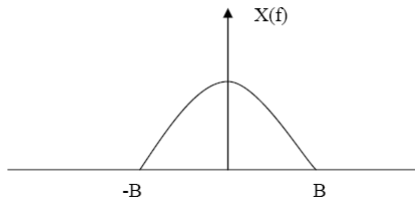


- Sampling = Original continuous signal $\times$ the Dirac comb function

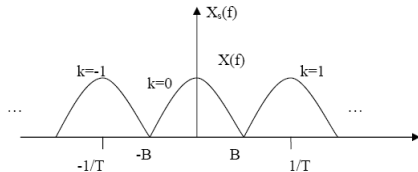
Effect of Sampling in the Frequency Domain

- Spectrum of original signal

Before Sampling:

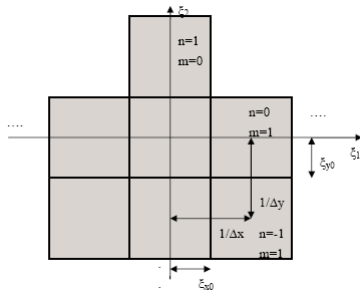
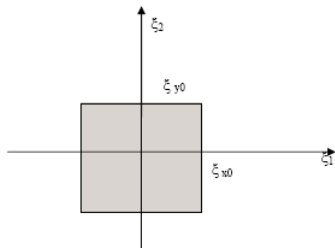


After Sampling:



2-D sampling

- Similar to 1-D sampling



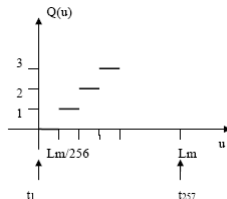
Quantization

- ▶ Why? Only limited bits (precision) available
- ▶ How? See the following example:
 - ▶ 8 bits (256 levels) to represent all possible values
 - ▶ A linear quantization scheme:

$$Q(u) = k \quad \text{if} \quad t_k \leq u < t_{k+1}$$

- ▶ Different quantization schemes based on:

1. Value Range
2. Value Distribution



Basic Mathematical Tools

- ▶ Element-wise vs. Matrix Operation
- ▶ Linear vs. Nonlinear Operation
- ▶ Arithmetic Operation
- ▶ Set and Logical Operation
- ▶ Spatial Operation
- ▶ Probability and Random Variables
- ▶ Vector and Matrix Operation

Element-wise vs. Matrix Operation

- ▶ Element-wise product

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} \\ a_{21}b_{21} & a_{22}b_{22} \end{bmatrix}$$

- ▶ Matrix product

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$

Linear vs. Nonlinear Operation

Consider a general operator, H , that produces an output image, $g(x, y)$, for a given input image, $f(x, y)$:

$$H[f(x, y)] = g(x, y)$$

H is said to be a linear operator if

$$\begin{aligned} H[a_i f_i(x, y) + a_j f_j(x, y)] &= a_i H[f_i(x, y)] + a_j H[f_j(x, y)] \\ &= a_i g_i(x, y) + a_j g_j(x, y) \end{aligned}$$

Arithmetic Operation

- Four arithmetic operations

$$s(x, y) = f(x, y) + g(x, y)$$

$$d(x, y) = f(x, y) - g(x, y)$$

$$p(x, y) = f(x, y) \times g(x, y)$$

$$v(x, y) = f(x, y) \div g(x, y)$$

(mean=0)
Average $\Rightarrow$ Add noise
Difference
Image of different styles
cloud $\times$ Apple
 $\Rightarrow$ Apple shape
cloud
surface

- Example: Noise reduction by averaging over noisy images.

Set and Logical Operation

- ▶ Basic set operations: subset, union, intersection, complement
...
- ▶ Logical operations: *Merge* AND, OR, NOT, XOR...
- ▶ Applications: Morphological processing

Examples!

*focus more details,
lose some amplitude!*

Spatial Operation

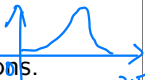
Board idea

月野卷 $\Rightarrow$ complement $\Rightarrow$ Black white reversed

- Single pixel operation.

Example: find the complement of image

low contrast



- Neighborhood Operations.

Example: Local averaging $\Rightarrow$ reduce the noise!

- Geometric Transformation

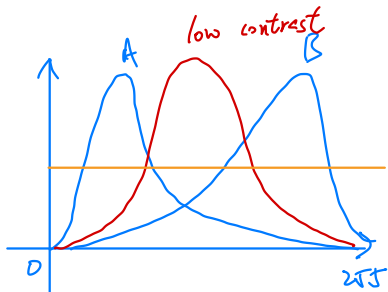
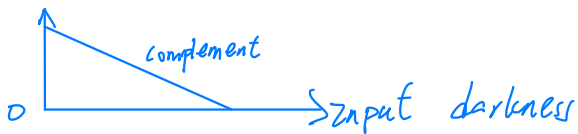
- Spatial transformation of coordinates

- Intensity interpolation

- Image Registration

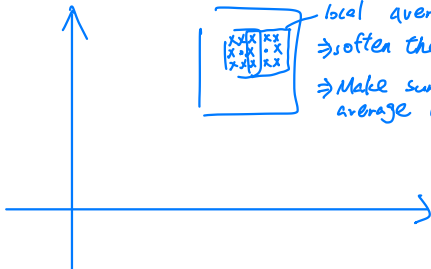


Transformation Name	Affine Matrix, T	Coordinate Equations	Example
Identity	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v$ $y = w$	
Scaling	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = c_x v$ $y = c_y w$	
Rotation	$\begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v \cos \theta - w \sin \theta$ $y = v \sin \theta + w \cos \theta$	
Translation	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ t_x & t_y & 1 \end{bmatrix}$	$x = v + t_x$ $y = w + t_y$	
Shear (vertical)	$\begin{bmatrix} 1 & 0 & 0 \\ s_v & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v + s_v w$ $y = w$	
Shear (horizontal)	$\begin{bmatrix} 1 & s_h & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v$ $y = s_h v + w$	



Don't know the details!
good contrast

1	0	0
0	1	0
0	0	1



local averaging
 ⇒ soften the edges?
 ⇒ Make sure that average
 average local noise

x	x	x
x	0	x
x	x	x

$$Pv = \lambda v$$

$$(P - \lambda I)v = 0$$

$$\begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = 0$$

$$\lambda^2 - 4\lambda + 4 - 1 = 0$$

$$\lambda^2 - 4\lambda + 3 = 0$$

$$(\lambda - 3)(\lambda - 1) = 0$$

$$\lambda = 3 \quad \text{or} \quad \lambda = 1$$

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$

↓

$$x + y = 0$$

$$x = -y$$

$$v_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \end{pmatrix}$$

↓

$$\begin{pmatrix} -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$-x + y = 0$$

$$x = y$$

$$\begin{pmatrix} 1 \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Probability and Random Variables

- ▶ Sample space, events and probability
- ▶ Conditional probability, Bayes' rule
- ▶ Independence, the law of total probability
- ▶ Random variables, Probability Mass Function, Probability Density Function
- ▶ Mean, variance, and higher order moments

Introduction of Matrix Algebra

- ▶ Indexing in matrices and images. Try this example in Matlab:

```
a = meshgrid([1:10]);  
a(1,2) = 100;  
figure; imagesc(a); grid on; axis image;  
figure; imagesc(a); grid on; axis xy;
```

Basic Rules and Definitions in Matrix Algebra

► Let

$$A = \{a[m, n]\} = \begin{bmatrix} a[1, 1] & a[1, 2] & \cdots & a[1, N] \\ a[2, 1] & & \ddots & \vdots \\ \vdots & & & \vdots \\ a[M, 1] & a[M, 2] & \cdots & a[M, N] \end{bmatrix}$$

► Transpose:

$$A^T = \{a[n, m]\},$$

$$\text{e.g. } A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Basic Rules and Definitions in Matrix Algebra

- Addition and multiplication:

$$C = A + B = \{a[m, n] + b[m, n]\}$$

$$C = A \cdot B = \left\{ \sum_{k=1}^k a[m, k] b[k, n] \right\}.$$

- Symmetric matrix

$$A^T = A, \quad \text{Symmetric} \quad \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix},$$

Basic Rules and Definitions in Matrix Algebra

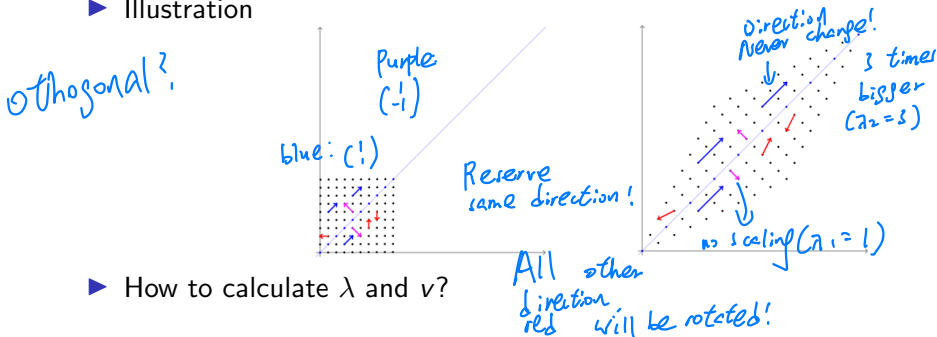
► Eigenvalues and Eigenvectors:

For a matrix P , if there is a scalar λ and a vector v such that:
 $Pv = \lambda v$, then λ : eigenvalue (characteristic value) and v :
eigenvector

$\Rightarrow$ Eigenvectors are linearly independent if they correspond to
distinct eigenvalues.

More on Eigenvalues and Eigenvectors

► Illustration



► How to calculate λ and v ?

$$Pv = \lambda v \Rightarrow |P - \lambda I| = 0.$$

► How to understand λ and v ?

- v : basis of the vector space.
- λ : weighting coefficient of the vector.

Reading Assignment

- ▶ Chapter 1 and 2 of the textbook

What is Transform?

- ▶ Convert the original signal from one domain to another
 - ▶ Transformed domain data have desirable properties
 - ▶ Transformed data is decorrelated. (Energy is redistributed among only a small set of transform coefficients.)
- ▶ Original data is obtained by applying inverse transform
- ▶ Transform could be : Continuous or Discrete
- ▶ Discrete Transform represented by a matrix
⇒ Forward transform

$$\vec{y} = T\vec{x}.$$

Scaling!

Basic Transform Operation

- ▶ X : Input sequence (of size N)
- ▶ T : Transformation matrix ($N \times N$)
- ▶ Y : Output sequence (of size N)
- ▶ Forward Transform : $Y = TX$
- ▶ Inverse Transform : $X = T^{-1}Y$ *if $\exists$ inverse operation!*
- ▶ T can be viewed as coordinate rotation matrix
- ▶ T^{-1} can be viewed as a set of basis elements

Two Views of Transforms

Rotate origin!
image!

Transform can be viewed as:

- ▶ Rotation of axes of the original data.
- ▶ Obtaining weights for the basis elements to express input in terms of basis elements

a typical example

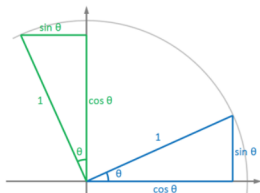
- 2D:

$$x' = x \cos \theta + y \sin \theta \quad y' = -x \sin \theta + y \cos \theta$$

rotation matrix

$$\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \cos \theta + y \sin \theta \\ -x \sin \theta + y \cos \theta \end{bmatrix}$$

NOTE: this is a 2 by 1 column vector,
 not a 2 by 2 matrix.



Rotation

Scaling

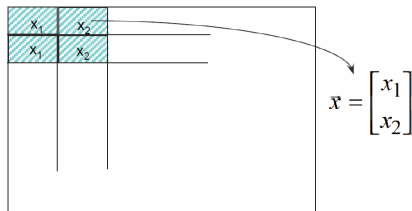
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Shearing and Reflection:

Dual pixel coding

2 pixels transform: divide the image into 1x2 blocks

- Note that typical transforms are 8x8 or 16x16



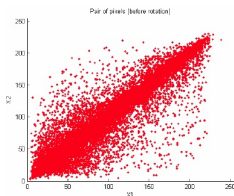
Transform as Rotation of Axes

- Let's look at an example.

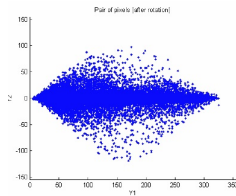
$$Y = \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \begin{bmatrix} \cos 45^\circ & \sin 45^\circ \\ -\sin 45^\circ & \cos 45^\circ \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$



Source Image: Lenna

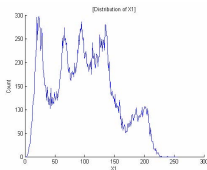


before rotation

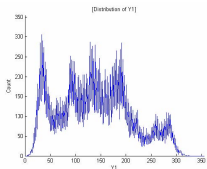


after rotation

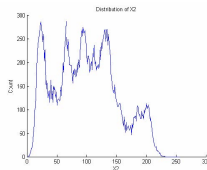
Transform as Rotation of Axes



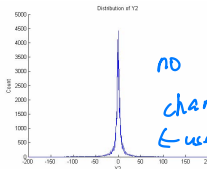
before rotation (x1)



after rotation (y1)



before rotation (x2)



after rotation (y2)

- ▶ Notice the variance of Y2 is smaller than the variance of X2.
- ▶ Compression: apply entropy coder on Y2

Effect of Axes Rotation

- ▶ Transform rotates the axes from x_1 - x_2 to y_1 - y_2
 - ▶ The energy in the original image is more or less equally distributed in both the x_1 and x_2 directions
 - ▶ the energy distribution along the y_1 and y_2 axes in the transformed data is highly skewed
 - ▶ More energy is packed along y_1
- ▶ This is an example of the energy packing property ^{use less bits} that is desirable in a transform. ^{just rotation}
- ▶ Transform operation itself does not achieve any compression;
- ▶ Transform operation aims at decorrelating the original data and compacting a large fraction to few transform components

Two Views of Transforms

Transform can be viewed as:

- ▶ Rotation of axes of the original data.
- ▶ Obtaining weights for the basis elements to express input in terms of basis elements

Transform as Basis Function Decomposition

Let $S = T^{-1}$ be the inverse transformation matrix, then

$$\vec{x} = S\vec{y} = [\vec{s}_1, \vec{s}_2, \dots, \vec{s}_N] \overset{\text{linear combination}}{\begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_N \end{bmatrix}} = \sum_i y_i \vec{s}_i$$

- Each y_i can be considered as weight for the basis element $\vec{s}_i$ to express input x in terms of basis elements.

Various Kinds of Transforms

Include:

- ▶ Discrete Fourier Transform (DFT)
- ▶ Fast Fourier Transform (FFT) is a fast computation Scheme for DFT
- ▶ Discrete Cosine Transform (DCT)
- ▶ Discrete Sine Transform
- ▶ Haar Transform
- ▶ Walsh-Hadamard Transform
- ▶ (Several Others)

Fourier Transform

- ▶ Original Definition: $f(t) \rightarrow F(\omega) = \int f(t)e^{-j\omega t} dt$
 - ▶ If t and ω are scalar variables, 1-D FT.
 - ▶ If t and ω are vector variables, n -D FT.
- ▶ For an $n \times n$ block of pixels, f , the forward 2-D DFT is defined as

$$F(u, v) = \frac{1}{n} \sum_{x=0}^{n-1} \sum_{y=0}^{n-1} f(x, y) \exp \left\{ -\frac{2\pi j(\overbrace{ux + vy}^{\text{horizontally}})}{n} \right\}$$

- ▶ What if we take discrete values, DFT in matrix view.

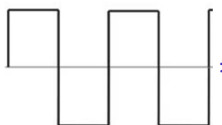
Fourier series reminder

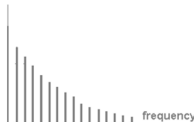
Example



$$f(x) = \sin x + \frac{1}{3} \sin 3x + \dots$$

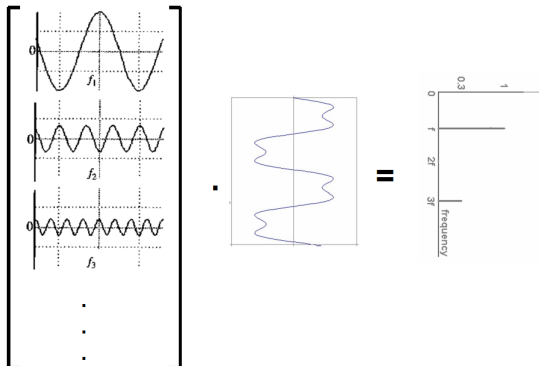
Fourier series for a square wave


$$f(x) = \sum_{n=1,3,5,\dots} \frac{1}{n} \sin nx$$



FT: Change of basis

$$M \quad f(x) = F(\omega)$$



IFT: Change of basis

$$M^{-1} F(\omega) = f(x)$$

